```
mysql> use Lab2;

Database changed

mysql> CREATE TABLE employees (emp_id INT PRIMARY KEY, first_name VARCHAR(50),
last_name VARCHAR(50), dept_id INT, salary DECIMAL(10,2), hire_date DATE, is_active
BOOLEAN);

Query OK, 0 rows affected (0.06 sec)


mysql> INSERT INTO employees (emp_id, first_name, last_name, dept_id, salary, hire_date,
is_active) VALUES (101, 'Alice', 'Smith', 10, 75000.00, '2021-03-15', TRUE), (102, 'Bob',
'Johnson', 20, 52000.00, '2022-06-01', TRUE), (103, 'Charlie', 'Brown', 10, 82000.00, '2019-
11-20', TRUE), (104, 'Diana', 'Prince', 30, 48000.00, '2023-01-10', FALSE), (105, 'Evan',
'Wright', 20, 61000.00, '2020-08-05', TRUE);

Query OK, 5 rows affected (0.01 sec)

Records: 5  Duplicates: 0  Warnings: 0


mysql> CREATE TABLE departments (dept_id INT PRIMARY KEY, dept_name VARCHAR(100),
location VARCHAR(100), budget DECIMAL(12,2));

Query OK, 0 rows affected (0.02 sec)


mysql> INSERT INTO departments (dept_id, dept_name, location, budget) VALUES (10,
'Engineering', 'Building A', 500000.00), (20, 'Marketing', 'Building B', 200000.00), (30,
'Human Resources', 'Building A', 150000.00), (40, 'Finance', 'Building C', 350000.00), (50,
'Research', 'Building D', 600000.00);

Query OK, 5 rows affected (0.01 sec)

Records: 5  Duplicates: 0  Warnings: 0


mysql> CREATE TABLE orders (order_id INT PRIMARY KEY, customer_name VARCHAR(100),
order_date DATE, total_amount DECIMAL(10,2), status VARCHAR(30));

Query OK, 0 rows affected (0.03 sec)


mysql> INSERT INTO orders (order_id, customer_name, order_date, total_amount, status)
VALUES (5001, 'TechCorp LLC', '2024-01-15', 1250.50, 'Completed'), (5002, 'Global Media',
```

'2024-01-18', 450.00, 'Pending'), (5003, 'Alpha Retail', '2024-01-20', 3100.00, 'Completed'), (5004, 'Beta Systems', '2024-01-22', 150.25, 'Cancelled'), (5005, 'Gamma Inc', '2024-01-25', 890.00, 'Shipped');

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM employees;

+--------+------------+-----------+---------+----------+------------+-----------+

| emp_id | first_name | last_name | dept_id | salary | hire_date | is_active |

+--------+------------+-----------+---------+----------+------------+-----------+

| 101 | Alice | Smith | 10 | 75000.00 | 2021-03-15 | 1 |

| 102 | Bob | Johnson | 20 | 52000.00 | 2022-06-01 | 1 |

| 103 | Charlie | Brown | 10 | 82000.00 | 2019-11-20 | 1 |

| 104 | Diana | Prince | 30 | 48000.00 | 2023-01-10 | 0 |

| 105 | Evan | Wright | 20 | 61000.00 | 2020-08-05 | 1 |

+--------+------------+-----------+---------+----------+------------+-----------+

5 rows in set (0.00 sec)

mysql> SELECT * FROM departments;

+---------+-----------------+------------+-----------+

| dept_id | dept_name | location | budget |

+---------+-----------------+------------+-----------+

| 10 | Engineering | Building A | 500000.00 |

| 20 | Marketing | Building B | 200000.00 |

| 30 | Human Resources | Building A | 150000.00 |

| 40 | Finance | Building C | 350000.00 |

| 50 | Research | Building D | 600000.00 |

+---------+-----------------+------------+-----------+

5 rows in set (0.01 sec)

```
mysql> SELECT * FROM orders;
+----------+---------------+------------+--------------+-----------+
| order_id | customer_name | order_date | total_amount | status    |
+----------+---------------+------------+--------------+-----------+
|     5001 | TechCorp LLC  | 2024-01-15 |      1250.50 | Completed |
|     5002 | Global Media  | 2024-01-18 |       450.00 | Pending   |
|     5003 | Alpha Retail  | 2024-01-20 |      3100.00 | Completed |
|     5004 | Beta Systems  | 2024-01-22 |       150.25 | Cancelled |
|     5005 | Gamma Inc     | 2024-01-25 |       890.00 | Shipped   |
+----------+---------------+------------+--------------+-----------+
5 rows in set (0.00 sec)

mysql> SHOW TABLES;
+----------------+
| Tables_in_lab2 |
+----------------+
| departments    |
| employees      |
| orders         |
+----------------+
3 rows in set (0.03 sec)

mysql> INSERT INTO employees (emp_id, first_name, last_name, dept_id, salary, hire_date,
is_active) VALUES (106, 'Fiona', 'Gallagher', 10, 65000.00, '2024-02-01', TRUE);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO departments (dept_id, dept_name, location, budget) VALUES (60,
'Quality Assurance', 'Building B', 250000.00);
```

Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO orders (order_id, customer_name, order_date, total_amount, status) VALUES (5006, 'Delta Retail', '2024-01-28', 2100.00, 'Pending');

Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM employees;

```
+--------+------------+-----------+---------+----------+------------+-----------+
| emp_id | first_name | last_name | dept_id | salary   | hire_date  | is_active |
+--------+------------+-----------+---------+----------+------------+-----------+
|    101 | Alice      | Smith     |      10 | 75000.00 | 2021-03-15 |         1 |
|    102 | Bob        | Johnson   |      20 | 52000.00 | 2022-06-01 |         1 |
|    103 | Charlie    | Brown     |      10 | 82000.00 | 2019-11-20 |         1 |
|    104 | Diana      | Prince    |      30 | 48000.00 | 2023-01-10 |         0 |
|    105 | Evan       | Wright    |      20 | 61000.00 | 2020-08-05 |         1 |
|    106 | Fiona      | Gallagher |      10 | 65000.00 | 2024-02-01 |         1 |
+--------+------------+-----------+---------+----------+------------+-----------+
```

6 rows in set (0.00 sec)

mysql> SELECT first_name, last_name, salary FROM employees;

```
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Alice      | Smith     | 75000.00 |
| Bob        | Johnson   | 52000.00 |
| Charlie    | Brown     | 82000.00 |
| Diana      | Prince    | 48000.00 |
| Evan       | Wright    | 61000.00 |
```

```
| Fiona      | Gallagher | 65000.00 |
+------------+-----------+----------+
6 rows in set (0.00 sec)

mysql> SELECT * FROM employees WHERE dept_id = 10;
+--------+------------+-----------+---------+----------+------------+-----------+
| emp_id | first_name | last_name | dept_id | salary   | hire_date  | is_active |
+--------+------------+-----------+---------+----------+------------+-----------+
|    101 | Alice      | Smith     |      10 | 75000.00 | 2021-03-15 |         1 |
|    103 | Charlie    | Brown     |      10 | 82000.00 | 2019-11-20 |         1 |
|    106 | Fiona      | Gallagher |      10 | 65000.00 | 2024-02-01 |         1 |
+--------+------------+-----------+---------+----------+------------+-----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM orders WHERE status = 'Completed';
+----------+---------------+------------+--------------+-----------+
| order_id | customer_name | order_date | total_amount | status    |
+----------+---------------+------------+--------------+-----------+
|     5001 | TechCorp LLC  | 2024-01-15 |      1250.50 | Completed |
|     5003 | Alpha Retail  | 2024-01-20 |      3100.00 | Completed |
+----------+---------------+------------+--------------+-----------+
2 rows in set (0.00 sec)

mysql> UPDATE employees SET salary = 55000.00 WHERE emp_id = 102;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE orders SET status = 'Shipped' WHERE order_id = 5002;
```

```
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE departments SET budget = budget + 20000.00 WHERE dept_id = 20;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> DELETE FROM orders WHERE order_id = 5004;
Query OK, 1 row affected (0.03 sec)

mysql> DELETE FROM employees WHERE emp_id = 104;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM employees WHERE salary > 60000.00;
+--------+------------+-----------+---------+----------+------------+-----------+
| emp_id | first_name | last_name | dept_id | salary   | hire_date  | is_active |
+--------+------------+-----------+---------+----------+------------+-----------+
|    101 | Alice      | Smith     |      10 | 75000.00 | 2021-03-15 |         1 |
|    103 | Charlie    | Brown     |      10 | 82000.00 | 2019-11-20 |         1 |
|    105 | Evan       | Wright    |      20 | 61000.00 | 2020-08-05 |         1 |
|    106 | Fiona      | Gallagher |      10 | 65000.00 | 2024-02-01 |         1 |
+--------+------------+-----------+---------+----------+------------+-----------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM orders WHERE order_date BETWEEN '2024-01-15' AND '2024-01-20';
+----------+---------------+------------+--------------+-----------+
| order_id | customer_name | order_date | total_amount | status    |
+----------+---------------+------------+--------------+-----------+
```

```
|    5001 | TechCorp LLC  | 2024-01-15 |     1250.50 | Completed |
|    5002 | Global Media  | 2024-01-18 |      450.00 | Shipped   |
|    5003 | Alpha Retail  | 2024-01-20 |     3100.00 | Completed |
+---------+---------------+------------+-------------+-----------+
3 rows in set (0.01 sec)

mysql> SELECT * FROM departments WHERE location IN ('Building A', 'Building B');
+---------+------------------+------------+-----------+
| dept_id | dept_name        | location   | budget    |
+---------+------------------+------------+-----------+
|      10 | Engineering      | Building A | 500000.00 |
|      20 | Marketing        | Building B | 220000.00 |
|      30 | Human Resources  | Building A | 150000.00 |
|      60 | Quality Assurance | Building B | 250000.00 |
+---------+------------------+------------+-----------+
4 rows in set (0.01 sec)

mysql> SELECT * FROM employees WHERE last_name LIKE 'S%';
+--------+------------+-----------+---------+----------+------------+-----------+
| emp_id | first_name | last_name | dept_id | salary   | hire_date  | is_active |
+--------+------------+-----------+---------+----------+------------+-----------+
|    101 | Alice      | Smith     |      10 | 75000.00 | 2021-03-15 |         1 |
+--------+------------+-----------+---------+----------+------------+-----------+
1 row in set (0.00 sec)

mysql> SELECT * FROM employees ORDER BY salary DESC;
+--------+------------+-----------+---------+----------+------------+-----------+
| emp_id | first_name | last_name | dept_id | salary   | hire_date  | is_active |
```

```
+--------+------------+-----------+---------+----------+------------+-----------+
|    103 | Charlie    | Brown     |      10 | 82000.00 | 2019-11-20 |         1 |
|    101 | Alice      | Smith     |      10 | 75000.00 | 2021-03-15 |         1 |
|    106 | Fiona      | Gallagher |      10 | 65000.00 | 2024-02-01 |         1 |
|    105 | Evan       | Wright    |      20 | 61000.00 | 2020-08-05 |         1 |
|    102 | Bob        | Johnson   |      20 | 55000.00 | 2022-06-01 |         1 |
+--------+------------+-----------+---------+----------+------------+-----------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM departments ORDER BY dept_name ASC;
+---------+------------------+------------+-----------+
| dept_id | dept_name        | location   | budget    |
+---------+------------------+------------+-----------+
|      10 | Engineering      | Building A | 500000.00 |
|      40 | Finance          | Building C | 350000.00 |
|      30 | Human Resources  | Building A | 150000.00 |
|      20 | Marketing        | Building B | 220000.00 |
|      60 | Quality Assurance | Building B | 250000.00 |
|      50 | Research         | Building D | 600000.00 |
+---------+------------------+------------+-----------+
6 rows in set (0.00 sec)

mysql> SELECT COUNT(*) AS total_employees FROM employees;
+-----------------+
| total_employees |
+-----------------+
|               5 |
+-----------------+
```

1 row in set (0.02 sec)

mysql> SELECT AVG(total_amount) AS average_order_amount FROM orders;

+----------------------+

| average_order_amount |

+----------------------+

| 1558.100000 |

+----------------------+

1 row in set (0.01 sec)

mysql> INSERT INTO employees (emp_id, first_name, last_name, dept_id, salary, hire_date, is_active) VALUES (107, 'George', 'Clark', 30, 50000.00, '2024-02-05', TRUE), (108, 'Hannah', 'Abbott', 10, 71000.00, '2024-02-10', TRUE);

Query OK, 2 rows affected (0.01 sec)

Records: 2 Duplicates: 0 Warnings: 0

mysql> UPDATE employees SET salary = salary * 1.10 WHERE dept_id = 10;

Query OK, 4 rows affected (0.01 sec)

Rows matched: 4 Changed: 4 Warnings: 0

mysql> UPDATE employees SET is_active = FALSE WHERE hire_date < '2021-01-01';

Query OK, 2 rows affected (0.01 sec)

Rows matched: 2 Changed: 2 Warnings: 0

mysql> SELECT * FROM employees WHERE is_active = TRUE AND salary > 60000.00;

+--------+------------+-----------+---------+----------+------------+-----------+

| emp_id | first_name | last_name | dept_id | salary | hire_date | is_active |

+--------+------------+-----------+---------+----------+------------+-----------+

| 101 | Alice | Smith | 10 | 82500.00 | 2021-03-15 | 1 |

```
|    106 | Fiona      | Gallagher |      10 | 71500.00 | 2024-02-01 |         1 |
|    108 | Hannah     | Abbott    |      10 | 78100.00 | 2024-02-10 |         1 |
+--------+------------+-----------+---------+----------+------------+-----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM orders WHERE status = 'Cancelled' OR total_amount > 2000.00;
+----------+---------------+------------+--------------+-----------+
| order_id | customer_name | order_date | total_amount | status    |
+----------+---------------+------------+--------------+-----------+
|     5003 | Alpha Retail  | 2024-01-20 |      3100.00 | Completed |
|     5006 | Delta Retail  | 2024-01-28 |      2100.00 | Pending   |
+----------+---------------+------------+--------------+-----------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM employees WHERE dept_id NOT IN (10, 30);
+--------+------------+-----------+---------+----------+------------+-----------+
| emp_id | first_name | last_name | dept_id | salary   | hire_date  | is_active |
+--------+------------+-----------+---------+----------+------------+-----------+
|    102 | Bob        | Johnson   |      20 | 55000.00 | 2022-06-01 |         1 |
|    105 | Evan       | Wright    |      20 | 61000.00 | 2020-08-05 |         0 |
+--------+------------+-----------+---------+----------+------------+-----------+
2 rows in set (0.01 sec)

mysql> SELECT * FROM employees WHERE LOWER(first_name) LIKE '%a%';
+--------+------------+-----------+---------+----------+------------+-----------+
| emp_id | first_name | last_name | dept_id | salary   | hire_date  | is_active |
+--------+------------+-----------+---------+----------+------------+-----------+
|    101 | Alice      | Smith     |      10 | 82500.00 | 2021-03-15 |         1 |
```

```
|    103 | Charlie    | Brown     |      10 | 90200.00 | 2019-11-20 |        0 |
|    105 | Evan       | Wright    |      20 | 61000.00 | 2020-08-05 |        0 |
|    106 | Fiona      | Gallagher |      10 | 71500.00 | 2024-02-01 |        1 |
|    108 | Hannah     | Abbott    |      10 | 78100.00 | 2024-02-10 |        1 |
+--------+------------+-----------+---------+----------+------------+-----------+
5 rows in set (0.01 sec)

mysql> SELECT dept_id, SUM(salary) AS total_salary FROM employees GROUP BY dept_id;
+---------+--------------+
| dept_id | total_salary |
+---------+--------------+
|      10 |    322300.00 |
|      20 |    116000.00 |
|      30 |     50000.00 |
+---------+--------------+
3 rows in set (0.01 sec)

mysql> SELECT dept_id, COUNT(*) AS total_employees FROM employees GROUP BY dept_id;
+---------+-----------------+
| dept_id | total_employees |
+---------+-----------------+
|      10 |               4 |
|      20 |               2 |
|      30 |               1 |
+---------+-----------------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT dept_id, SUM(salary) AS total_salary FROM employees GROUP BY dept_id HAVING SUM(salary) > 100000.00;
+---------+--------------+
| dept_id | total_salary |
+---------+--------------+
|      10 |    322300.00 |
|      20 |    116000.00 |
+---------+--------------+
2 rows in set (0.01 sec)

mysql> SELECT status, MAX(total_amount) AS maximum_amount, MIN(total_amount) AS minimum_amount FROM orders GROUP BY status;
+-----------+----------------+----------------+
| status    | maximum_amount | minimum_amount |
+-----------+----------------+----------------+
| Completed |       3100.00 |       1250.50 |
| Shipped   |        890.00 |        450.00 |
| Pending   |       2100.00 |       2100.00 |
+-----------+----------------+----------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);
+--------+------------+-----------+---------+----------+------------+-----------+
| emp_id | first_name | last_name | dept_id | salary   | hire_date  | is_active |
+--------+------------+-----------+---------+----------+------------+-----------+
|    101 | Alice      | Smith     |      10 | 82500.00 | 2021-03-15 |         1 |
|    103 | Charlie    | Brown     |      10 | 90200.00 | 2019-11-20 |         0 |
|    106 | Fiona      | Gallagher |      10 | 71500.00 | 2024-02-01 |         1 |
|    108 | Hannah     | Abbott    |      10 | 78100.00 | 2024-02-10 |         1 |
```

```
+--------+------------+-----------+---------+----------+------------+-----------+
```

4 rows in set (0.00 sec)

```
mysql> SELECT * FROM employees WHERE dept_id IN (SELECT dept_id FROM departments
WHERE location = 'Building A');
+--------+------------+-----------+---------+----------+------------+-----------+
| emp_id | first_name | last_name | dept_id | salary   | hire_date  | is_active |
+--------+------------+-----------+---------+----------+------------+-----------+
|    101 | Alice      | Smith     |      10 | 82500.00 | 2021-03-15 |         1 |
|    103 | Charlie    | Brown     |      10 | 90200.00 | 2019-11-20 |         0 |
|    106 | Fiona      | Gallagher |      10 | 71500.00 | 2024-02-01 |         1 |
|    107 | George     | Clark     |      30 | 50000.00 | 2024-02-05 |         1 |
|    108 | Hannah     | Abbott    |      10 | 78100.00 | 2024-02-10 |         1 |
+--------+------------+-----------+---------+----------+------------+-----------+
```

5 rows in set (0.00 sec)

```
mysql> DELETE FROM orders WHERE customer_name IN (SELECT customer_name FROM
(SELECT customer_name FROM orders WHERE status = 'Cancelled') AS
cancelled_customers);
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM employees WHERE last_name IS NULL;
```

Empty set (0.00 sec)

```
mysql> SELECT CONCAT(first_name, ' ', last_name) AS full_name FROM employees;
+-----------------+
| full_name       |
+-----------------+
| Alice Smith     |
```

```
| Bob Johnson     |
| Charlie Brown   |
| Evan Wright     |
| Fiona Gallagher |
| George Clark    |
| Hannah Abbott   |
+-----------------+
7 rows in set (0.00 sec)

mysql> SELECT * FROM employees ORDER BY salary DESC LIMIT 3;
+--------+------------+-----------+---------+----------+------------+-----------+
| emp_id | first_name | last_name | dept_id | salary   | hire_date  | is_active |
+--------+------------+-----------+---------+----------+------------+-----------+
|    103 | Charlie    | Brown     |      10 | 90200.00 | 2019-11-20 |         0 |
|    101 | Alice      | Smith     |      10 | 82500.00 | 2021-03-15 |         1 |
|    108 | Hannah     | Abbott    |      10 | 78100.00 | 2024-02-10 |         1 |
+--------+------------+-----------+---------+----------+------------+-----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM employees ORDER BY dept_id ASC, salary DESC;
+--------+------------+-----------+---------+----------+------------+-----------+
| emp_id | first_name | last_name | dept_id | salary   | hire_date  | is_active |
+--------+------------+-----------+---------+----------+------------+-----------+
|    103 | Charlie    | Brown     |      10 | 90200.00 | 2019-11-20 |         0 |
|    101 | Alice      | Smith     |      10 | 82500.00 | 2021-03-15 |         1 |
|    108 | Hannah     | Abbott    |      10 | 78100.00 | 2024-02-10 |         1 |
|    106 | Fiona      | Gallagher |      10 | 71500.00 | 2024-02-01 |         1 |
|    105 | Evan       | Wright    |      20 | 61000.00 | 2020-08-05 |         0 |
```

```
|    102 | Bob        | Johnson    |      20 | 55000.00 | 2022-06-01 |         1 |
|    107 | George     | Clark      |      30 | 50000.00 | 2024-02-05 |         1 |
+--------+------------+------------+---------+----------+------------+-----------+
7 rows in set (0.00 sec)

mysql> DELETE FROM departments WHERE budget < 200000.00;
Query OK, 1 row affected (0.01 sec)

mysql> UPDATE orders SET status = 'Cancelled' WHERE total_amount < 200.00;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> UPDATE employees SET salary = salary * 1.15 WHERE dept_id = (SELECT dept_id
FROM departments WHERE budget = (SELECT MAX(budget) FROM departments));
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> SELECT dept_id, AVG(salary) AS average_salary FROM employees GROUP BY dept_id
HAVING AVG(salary) > (SELECT AVG(salary) FROM employees);
+---------+----------------+
| dept_id | average_salary |
+---------+----------------+
|      10 |   80575.000000 |
+---------+----------------+
1 row in set (0.01 sec)

mysql> SELECT * FROM employees WHERE dept_id IN (SELECT dept_id FROM departments
WHERE budget > 300000.00);
+--------+------------+-----------+---------+----------+------------+-----------+
```

| emp_id | first_name | last_name | dept_id | salary | hire_date | is_active |

+--------+------------+-----------+---------+----------+------------+-----------+

| 101 | Alice | Smith | 10 | 82500.00 | 2021-03-15 | 1 |

| 103 | Charlie | Brown | 10 | 90200.00 | 2019-11-20 | 0 |

| 106 | Fiona | Gallagher | 10 | 71500.00 | 2024-02-01 | 1 |

| 108 | Hannah | Abbott | 10 | 78100.00 | 2024-02-10 | 1 |

+--------+------------+-----------+---------+----------+------------+-----------+

4 rows in set (0.00 sec)

mysql> SELECT DISTINCT salary FROM employees ORDER BY salary DESC LIMIT 1 OFFSET 2;

+----------+

| salary |

+----------+

| 78100.00 |

+----------+

1 row in set (0.00 sec)

mysql> UPDATE employees SET salary = CASE WHEN salary < 60000 THEN salary * 1.10 ELSE salary * 1.05 END;

Query OK, 7 rows affected (0.01 sec)

Rows matched: 7 Changed: 7 Warnings: 0

mysql> DELETE FROM employees e WHERE e.is_active = FALSE AND e.salary < (SELECT AVG(e2.salary) FROM employees e2 WHERE e2.dept_id = e.dept_id);

ERROR 1093 (HY000): You can't specify target table 'e' for update in FROM clause

mysql> SELECT d.dept_name, (SELECT COUNT(*) FROM employees e WHERE e.dept_id = d.dept_id) AS total_employees FROM departments d;

+------------------+-----------------+

| dept_name | total_employees |

```
+------------------+-----------------+
| Engineering      |               4 |
| Marketing        |               2 |
| Finance          |               0 |
| Research         |               0 |
| Quality Assurance |              0 |
+------------------+-----------------+
5 rows in set (0.00 sec)

mysql> SELECT customer_name FROM orders GROUP BY customer_name HAVING
COUNT(order_id) > 1 OR SUM(total_amount) > 1000.00;
+---------------+
| customer_name |
+---------------+
| TechCorp LLC  |
| Alpha Retail  |
| Delta Retail  |
+---------------+
3 rows in set (0.01 sec)

mysql> INSERT INTO employees (emp_id, first_name, last_name, dept_id, salary, hire_date,
is_active) SELECT emp_id + 500, first_name, last_name, 50, salary, hire_date, is_active
FROM employees WHERE dept_id = 10;
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> UPDATE orders SET status = CASE WHEN total_amount > 1000 THEN 'Completed'
WHEN total_amount BETWEEN 500 AND 1000 THEN 'Shipped' ELSE 'Pending' END;
Query OK, 2 rows affected (0.01 sec)
```

Rows matched: 5 Changed: 2 Warnings: 0